

TABLE I. DRY SATURATED MEDIA EXPERIMENT

Media	Dry Volume (ml)			Dry		
	Cocopeat	Husk	Total	Weight (g)	VWC %	Volume: Weight
Cocopeat: 393 Husk: 132	393	132	525	94	0	5.59

TABLE II. WET SATURATED MEDIA EXPERIMENT

Media	Dry Volume (ml)			Dry		
	Cocopeat	Husk	Total	Weight (g)	Water Content % (Formula)	Water Content % (MEC-10)
Cocopeat: 393 Husk: 132	393	132	525	404	59%	56%

TABLE III. MEC-10 SENSOR ERROR PERCENTAGE

Amount of Water	Reference Data (%)	MEC10 Experiment Data (%)	Percent Error % (MEC-10)
10 ml	5%	3.55%	29%
20 ml	9%	8%	11.11%
30 ml	13%	10%	23.10%
40 ml	18%	19%	5.55%
50 ml	23%	21.96%	4.52%
60 ml	28%	29%	3.57%
70 ml	33%	34.93%	5.85%
80 ml	38%	39%	2.63%
90 ml	43%	45.04%	4.74%

100 ml	48%	49%	2.08%
110 ml	53%	54.01%	1.09%
120 ml	56%	55.06%	5.07%

TABLE IV. CAPACITIVE SENSOR ERROR PERCENTAGE

Amount of Water	Reference Data (%)	Capacitive Sensor Experiment Data (%)	Percent Error % (Capacitive Sensor)
10 ml	5%	7.14%	42.80%
20 ml	9%	14.97%	66.33%
30 ml	13%	18.9%	45.38%
40 ml	18%	32.67%	81.50%
50 ml	23%	63.36%	175.48 %
60 ml	28%	82.27%	193.82 %
70 ml	33%	98.15%	158.29 %
80 ml	38%	96.38%	153.63 %
90 ml	43%	94.78%	120.42 %
100 ml	48%	100.19%	108.73 %
110 ml	53%	95.10%	79.43%
120 ml	56%	94.24%	65.33%

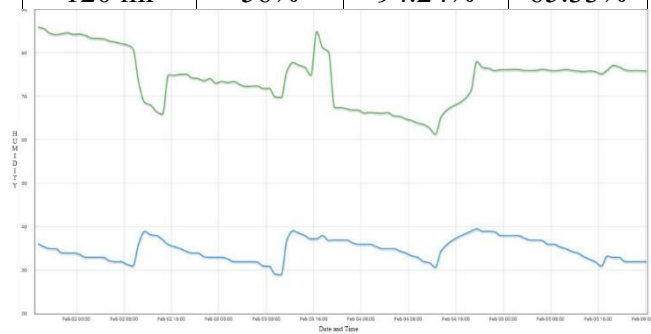


Fig. 1. Dashboard for automatic watering

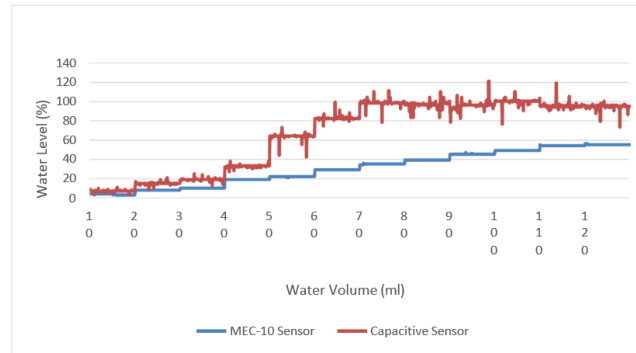


Fig. 2. Water level in planting media (for experiment using 120 ml water)

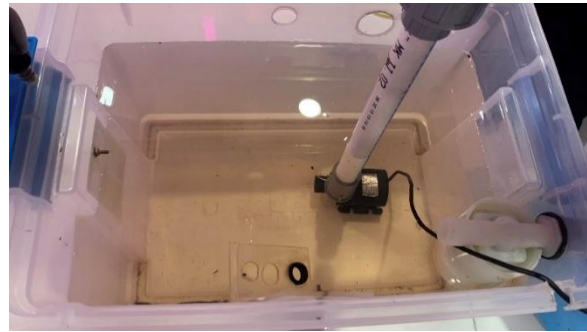


Fig. 3. DC water pump



Fig. 4. Front view of system